A Hierarchical Dissection of Multiscale Forcing on the Springtime Mesoscale Convective Systems in the United States

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ABSTRACT: Mesoscale convective systems (MCSs) play a key role in regulating variability in the U.S. water and energy cycle. Here a hierarchical dissection of the multiscale forcing of springtime MCSs is carried out through a two-step classification process. Hierarchical clustering is first applied to spatiotemporally evolving upper-tropospheric height fields to reveal large-scale forcing patterns of MCSs. Five distinct forcing patterns (clusters) are identified with three being "remotely forced" and two associated with "local growth." The upper-level troughs associated with these forcing patterns create broad envelopes downstream within which large-scale ascent and MCS genesis tend to occur. Further classification of MCSs based on MCS track locations reveals that local dynamic and thermodynamic forcing determines the precise locations of MCS genesis in the envelope created by large-scale forcing. Specifically, MCSs often occur near surface fronts in warm sectors of surface low pressure systems and are accompanied by low-level kinematic and moisture convergence driven by low-level jets (LLJs). Nearly 50% of spring MCSs are associated with potential instability realized through frontal lifting, and the highest probability of MCS genesis is seen with an environmental CAPE of ~1400 J kg^{-1} and CIN of ~150 J kg^{-1}. The positive trend of the U.S. MCS genesis frequency observed in recent decades is found to be driven by the cluster of MCSs forced at large scale by the Pacific storm track. Regression analysis further suggests that the growing phase of the Pacific decadal oscillation (PDO) modulates the associated MCS large-scale forcing and is ultimately responsible for the positive MCS trend.

SIGNIFICANCE STATEMENT: The purpose of this study is to provide a systematic classification of multiscale forcing factors triggering mesoscale convective system development over the United States. These storms are very active in spring and often lead to intense rainfall and other weather hazards such as lightning, hail, and tornadoes. They play a key role in the U.S. hydrological cycle and have been occurring more frequently over the past several decades. Our study reveals the detailed characteristics of atmospheric forcing leading to these storms. Such information lays theoretical grounds for designing prediction schemes of warm season severe weather and provides guidance for model development to improve climate models' simulation and long-term projection of these storms.

KEYWORDS: Atmospheric circulation; Convective storms/systems; Hydrologic cycle

1. Introduction

Mesoscale convective systems (MCSs), one of the largest forms of midlatitude convective storms, develop and mature when aggregated cumulonimbus clouds become organized and produce distinct mesoscale circulations (Houze 2004, 2018). MCSs are commonly observed in the tropics (Nesbitt et al. 2006; Yuan and Houze 2010) and over the midlatitude continents in eminent baroclinic zones (Laing and Fritsch 1997, 2000). Characterized by both regions of convective and stratiform precipitation (Houze 2004), MCSs account for a large proportion of precipitation in both the tropics and midlatitudes (Feng et al. 2019; Nesbitt et al. 2006).

In the United States, east of the Rocky Mountains, MCSs are very active and often generate severe weather in the boreal warm season (March–August) including strong straight-line winds, hail, and tornadoes (Bentley and Sparks 2003; Houze et al. 1990; Smith et al. 2012). They contribute 50%–70% of the total rainfall over the southern Great Plains in spring (Feng et al. 2016, 2019; Haberlie and Ashley 2019), and 60%–75% of extreme rainfall events in this region (Schumacher and Johnson 2006). Long-lasting, slow-moving MCSs are a major cause of flooding (Doswell et al. 1996; Kunkel et al. 2012; Schumacher and Johnson 2006; Stevenson and Schumacher 2014). In the past few decades, MCSs over the U.S. Great Plains have been occurring more frequently, leading to more intense precipitation (Feng et al. 2016). Their numbers and intensities are projected to grow further in a warming climate (Prein et al. 2017).

Reproducing MCSs and their associated precipitation in terms of variability and long-term trends remains a major challenge for climate models. The issue of MCS simulation in models is partly attributed to a long-standing warm and dry bias in the central United States during the warm season (Lin et al. 2017; Morcrette et al. 2018; Van Weverberg et al. 2018). In recent years, convection-permitting models (CPMs) displayed some promises in better capturing MCS precipitation structures (Cheruy et al. 2014). CPMs explicitly resolve rather than parameterize convection, which has enabled more skillful simulations of precipitation (Gao et al. 2017). Some studies have shown, however, that CPMs still have substantial dry biases and consequently underestimate MCS frequency over

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the central United States (Feng et al. 2018; Gao et al. 2017; Prein et al. 2020). It is also unclear if global models can reproduce all the large-scale environmental conditions and thermodynamic elements associated with springtime MCSs (Song et al. 2019).

A faithful representation of MCSs in a model hinges upon the model's capability of accurately simulating multiscale processes associated with MCS initiation and intensification. Maddox (1983) found that intense MCSs are often triggered by lifting motions ahead of large-scale troughs. Coniglio et al. (2010) linked the development speed and lifetime of MCSs with local dynamic and thermodynamic elements including the low-level jet (LLJ), atmospheric instability, frontal lifting, and vertical wind shear. Song et al. (2019) summarized several types of environments favoring the development of springtime MCSs. Most of those environments contain frontal systems, enhanced LLJs, large positive anomalies in convective available potential energy (CAPE), and negative anomalies in convective inhibition (CIN). Feng et al. (2019) divided the continental U.S. MCSs east of the Rocky Mountains into four separate geographical zones and discussed the large-scale and thermodynamic environments attendant with MCSs in each zone. The multiscale nature of MCSs is also intrinsically tied to the fact that MCSs can produce significant feedback to large-scale circulations through generating and injecting into the atmosphere new potential vorticity (Chen and Frank 1993; Feng et al. 2019; Fritsch et al. 1994; Yang et al. 2017).

Despite all the efforts to document observed characteristics of the U.S. MCSs, a detailed depiction of MCSs from the angle of the associated multiscale forcing has yet to emerge. Delineating the dynamical pathway connecting planetary-scale, synoptic-scale circulation anomalies to the actual MCS genesis and the mesoscale features accompanying its growth sets the process-level foundation for understanding the MCS climatology, variability, and long-term trends in a warming climate. This dynamical pathway, once identified, will also help model developers trace MCS biases to existing model errors in either local or remote factors (e.g., sea surface temperature biases). The cornerstone of constructing this pathway is a "hierarchical dissection" of the MCS-related forcing (i.e., partitioning the MCS-related forcing mechanisms into two hierarchical levels: level 1 identifies multiple distinct large-scale backgrounds of MCSs while level 2, for MCSs of similar large-scale backgrounds, identifies local dynamic and thermodynamic factors leading to different morphology of MCS tracks). It is worth noting that the "forcing" identified in this paper is mostly not the direct forcing of the MCS genesis (e.g., background ascent) but the "atmospheric features" (e.g., troughs and fronts) that impose the forcing. Nevertheless, we keep using the term "forcing" for the large-scale and local patterns identified in this paper to echo the nature of MCS-triggering mechanisms (e.g., rising motions driven by large-scale circulation patterns and local dynamic/thermodynamic factors).

Here, we conduct a hierarchical dissection of the multiscale forcing of the 2000–20 springtime MCSs in the United States. The analysis begins with the identification of distinct hemispheric-scale circulation patterns leading to MCS genesis over the United States via hierarchical clustering, followed by subclassification of MCS events (with similar large-scale forcing) according to MCS track locations. The dynamic and thermodynamic varieties in MCS initiations are discussed in terms of the different combinations of large-scale forcing patterns (e.g., synoptic-scale disturbances and jets) and local dynamic/thermodynamic elements (e.g., fronts, LLJs, instability, CAPE, and CIN), based upon the results of clustering. Springtime is the focus of this study because the rainfall is maximized during that time over the central Great Plains, where MCSs are a primary concern in contributing to extreme rainfall and flooding. Additionally, as a transitional season, spring is characterized by MCSs influenced by both distinct extratropical synoptic-scale disturbances and local dynamics/thermodynamics, providing many events suitable for a "hierarchical dissection." On the contrary, summer MCSs are more affected by local factors with synoptic-scale disturbances shifted to higher latitudes.

Following the introduction, this paper is organized as follows: Section 2 describes the data and methods for MCS tracking and clustering analysis; section 3 presents the main findings, highlighting the multiscale forcing patterns of U.S. MCSs in boreal spring and discussing the potential causes of interannual variability and long-term trends of MCSs; and section 4 provides some concluding remarks.

2. Data and methods

a. Data

MCSs in the present study are identified with the NCEP–CPC merged geostationary satellite infrared brightness temperature T_b used by Feng et al. (2018). This dataset merges all the available geostationary satellites and includes an approximately 4-km spatial resolution and 30-min temporal resolution (Janowiak et al. 2001). This T_b dataset has been proven to suit quantitative analyses across regional to global scales, such as the tracking of deep convective systems (Feng et al. 2018). This study also uses the fifth generation of the ECMWF atmospheric reanalysis data (ERA5) (Hersbach et al. 2020), with a horizontal resolution of $0.25° \times 0.25°$ (~30 km). The selected variables on pressure levels include geopotential, zonal and meridional winds, temperature, and specific humidity. The thermodynamic analysis also makes use of the ERA5 CAPE and CIN datasets. (Note that the ERA5 CAPE used in this study is the most unstable CAPE, rather than the surface-based CAPE.) The CPC global gridded daily precipitation dataset ($0.5° \times 0.5°$) (Xie et al. 2007) is applied to estimate the MCS precipitation and its contribution to the springtime average. In addition, we have also utilized the COBE dataset ($1° \times 1°$) for monthly-averaged sea surface temperature (SST) in our regression analysis (Ishii et al. 2005).

b. Tracking of MCSs

This study applies the "detect and spread" approach to capture continuous convective clouds of each time step (Boer and Ramanathan 1997; Feng et al. 2018; Futyan and Del Genio 2007) using the T_b dataset of 30-min time resolution. The idea of this approach is to identify "cold cores" below a certain T_b threshold as individual cloud centers and to spatially

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spread each cloud until a warmer threshold is met. Depending on different research purposes and datasets used, the thresholds vary significantly among different detection methods (Boer and Ramanathan 1997; Feng et al. 2012, 2018; Futyan and Del Genio 2007; Haberlie and Ashley 2019; Jirak et al. 2003; Williams and Houze 1987; Yuan and Houze 2010). For example, Rodgers et al. (1985) picked 221 K as the colder threshold and 251 K as the secondary threshold. In the Flexible Object Tracker (FLEXTRKR) developed by Feng et al. (2018), the cold core temperature was set below 225 K (smoothed by 40 km × 40 km pixels) and the cloud area expanded until the unsmoothed T_b reached 241 K.

Our detection method defines cold cores as continuous areas with $T_b < 221$ K. Each cold core expands until T_b reaches 231 K. The warm threshold in our method is lower than that in the previous two approaches because we intend to emphasize distinct and intense MCS events. If the entire cold cloud area ($T_b < 231$ K) exceeds 50 000 km^2, it will be pinpointed as a cold cloud system (CCS). The connection method between CCSs at adjacent time steps is similar to what was described by Williams and Houze (1987) and Futyan and Del Genio (2007). If, within two successive T_b images, a CCS overlaps itself by more than 50% going *either* forward or backward in time, this CCS is considered as the same system and the track continues. An MCS is defined as a CCS lasting at least 6 h. Once the detection finishes, all MCS tracks go through a postprocessing step that merges the adjacent tracks and cloud areas if the average distance between two tracks is less than 100 km within their overlapping time. Next, we eliminate the tracks shorter than 500 km. These two postprocessing steps intend to avoid unnecessary overcounting of MCSs and to focus on long-lived MCSs only. As seen from the results to be discussed, our method works well in reproducing MCS statistics and distributions that have been documented in previous studies.

c. *Clustering and subclustering of MCS events*

This study performs a first-level classification based on the large-scale forcing patterns triggering MCSs, followed by a second-level subclassification of MCS events according to MCS track locations. The hierarchical clustering algorithm adopted here is a "bottom-up" approach (agglomerative). The clustering starts with a list of "objects," which are vectors constructed from the observational datasets (e.g., geopotential height anomalies of four consecutive days prior to MCS geneses in the first level, or MCS track locations in the second level, as discussed later). At the beginning, each "object" is considered as an individual "cluster," and pairs of similar clusters are merged at the following steps depending on the mutual "distances" of clusters.

We employ the Ward's minimum variance clustering criterion (Park et al. 2014; Ward 1963; Zhao et al. 2017a,b) to compute the "distance" and merge the clusters. Ward's distance between two clusters (e.g., clusters p and q) is the difference between the error sum of squares (ESS) of the newly formed cluster pq (if p and q are to be merged) and the sum of the two clusters' ESS. The ESS for an individual cluster $\mathbf{P}$ is defined as

$$\mathrm{ESS} = \sum_{i=1}^{n} |\mathbf{P}_i - \overline{\mathbf{P}}|^2, \tag{1}$$

where n is the total number of objects (vectors) in a cluster and $\overline{\mathbf{P}}$ is the "center" (or "mean") of the cluster. Ward's distance between clusters p and q is thus expressed as follows:

$$d_{p,q} = \mathrm{ESS}_{pq} - \mathrm{ESS}_p - \mathrm{ESS}_q. \tag{2}$$

The first merging occurs when two clusters have the smallest Ward's distance. The following steps repeat the same procedure (i.e., computing Ward's distance for all pairs of clusters, finding the smallest distance, and merging) for the updated group of clusters. As this process continues, there will be one step where a significant "jump" or "increase" in the smallest Ward's distance happens. This indicates that the two clusters merged at this step have very different structures and thus should not be merged any further. The clustering analysis is thus terminated at the previous step and the number of clusters at the previous step is the final number of clusters. Each final cluster contains an ensemble of similar objects. Field variables corresponding to the objects within each final cluster are then averaged to produce composite patterns for each cluster.

Instead of classifying the large-scale forcing of MCSs with spatial patterns only, our approach uniquely identifies circulation patterns of different spatiotemporal characteristics. Specifically, the first-level clustering analysis regards the classification "objects" as the normalized daily geopotential height anomalies at 500 hPa within 20°–70°N, 150°E–80°W, from day −3 to day 0 of all the detected MCS events (day 0 being the calendar date enclosing the moment 12 h prior to the MCS genesis). The daily anomalies are obtained based on the 2000–20 daily climatological mean values. For each field of daily geopotential height anomalies from day −3 to day 0 of an MCS, the anomaly at every grid point is normalized by the maximum absolute value across the entire domain on that day and weighted by the cosine of the latitude of the corresponding grid point. Elements of the normalized, weighted matrixes are then realigned into vectors and the vectors from day −3 to day 0 are concatenated into a single vector. This single vector contains key spatiotemporal structure of geopotential height anomalies over four consecutive days leading up to the initiation of an MCS and becomes the classification object of the first-level hierarchical clustering. This first-level clustering yields five clusters in the end, with each cluster containing an ensemble of geopotential height anomalies with similar structures. We subsequently take an average (i.e., composite) of the geopotential height anomalies in each cluster to obtain five spatiotemporally unique large-scale forcing patterns.

The actual MCS genesis locations within each of these five clusters, however, remain quite diverse, suggesting additional constraints of MCS genesis from local dynamic and thermodynamic factors. We thus conduct the second-level clustering for MCS events in each of the five main clusters. The basis of the second-level clustering is the location of an MCS event.

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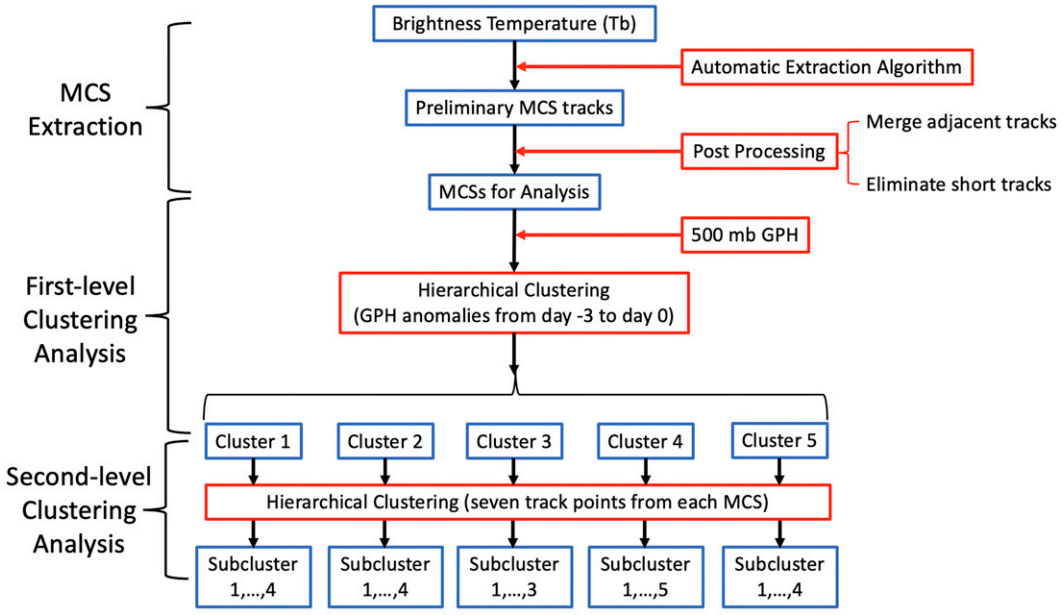


FIG. 1. Key steps of MCS detection, first-level clustering (based on large-scale pattern), and second-level clustering (based on MCS track morphology).

Specifically, the MCS track (i.e., latitudes and longitudes of the center of an MCS throughout its lifetime) is divided into six equal time segments. The latitudes and longitudes of the seven track points (a total of fourteen elements) associated with the six segments form a vector that becomes the "object" of the classification in the second-level clustering. This step of clustering is performed for each main cluster from the first level and the obtained clusters are called subclusters. The resultant number of subclusters in each of the five main clusters is 4, 4, 3, 5, and 4, respectively. Composite analysis based upon these subclusters is then carried out to reveal the additional roles played by local dynamic (e.g., fronts, LLJs) and thermodynamic (atmospheric stability) factors in modulating the precise life cycle characteristics of MCSs, including their genesis locations and tracking patterns. A flowchart that summarizes the key steps of MCS extraction, the first- and the second-level clustering analysis, is provided in Fig. 1.

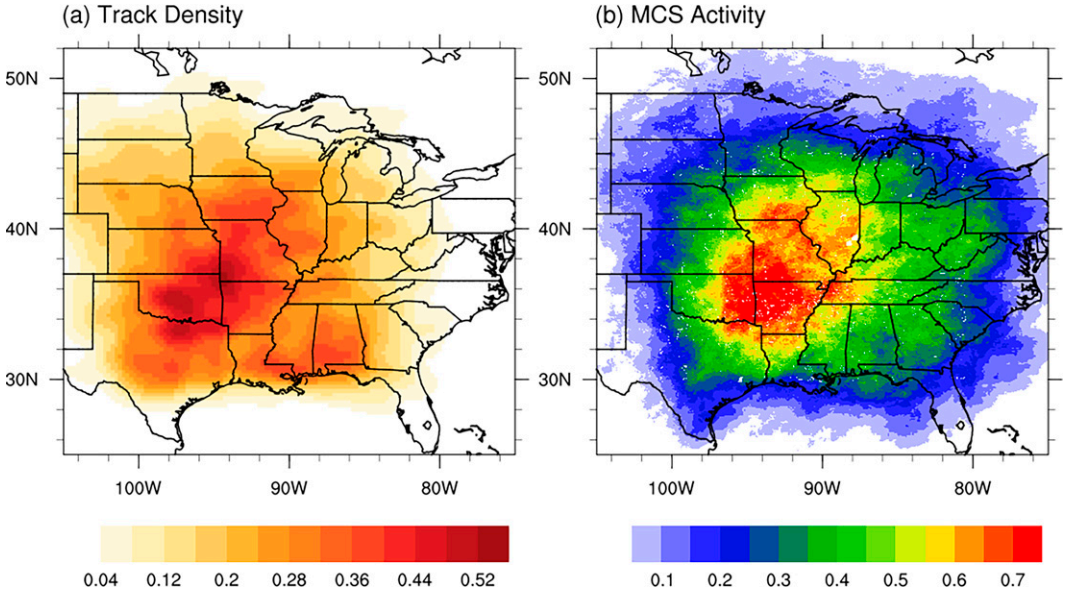


FIG. 2. (a) The climatological MCS track density and (b) MCS activity (intensity) (K) in boreal spring.

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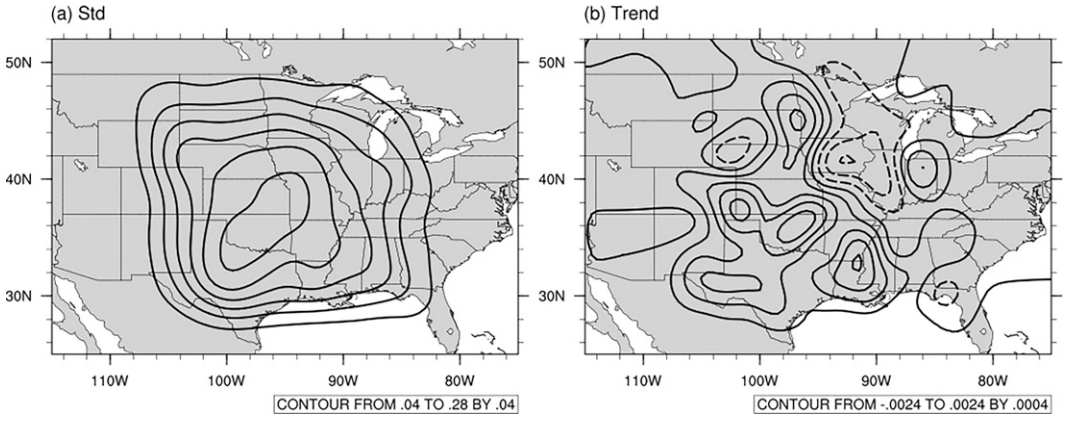


FIG. 3. (a) The interannual standard deviation and (b) trend of springtime MCS genesis frequency from 2000 to 2020.

3. Results

a. *MCS climatology*

The MCS detection domain of this study covers most of the central United States (30°–50°N, 80°–105°W). A total number of 1677 long-lived and intense MCSs in spring are tracked from 2000 to 2020, averaging 80 events per spring. This number is larger than that in Feng et al. (2019), which extracted about 63 events per spring with FLEXTRKR, and much smaller than the number of 171 events from Haberlie and Ashley (2019) detected from radar data. In fact, the detection number of MCSs varies significantly depending on the datasets and thresholds of the detection method (Boer and Ramanathan 1997; Feng et al. 2012, 2018; Futyan and Del Genio 2007; Haberlie and Ashley 2019; Jirak et al. 2003). The absolute number of MCSs, however, matters less compared to some more generally agreed findings from past studies such as the MCS contribution to the total rainfall in the United States. Specifically, detections based on satellite data often lead to fewer MCS events than that based on radar data, but they both agree on the result of MCS contribution to the springtime precipitation over the central United States (Feng et al. 2019; Haberlie and Ashley 2019). Feng et al. (2019) have further shown that the "robust MCSs" captured with satellite data are sufficient to cover the vast majority of the MCS precipitation. In addition, the merging of small convective cores in satellite-data-based MCS detection does not qualitatively affect the composites of multiscale forcing either since those convective cores share the same synoptic- and mesoscale environments with the merged cloud cluster.

Figure 2 shows the spatial distributions of the climatological MCS track density and activity in spring. For the track density

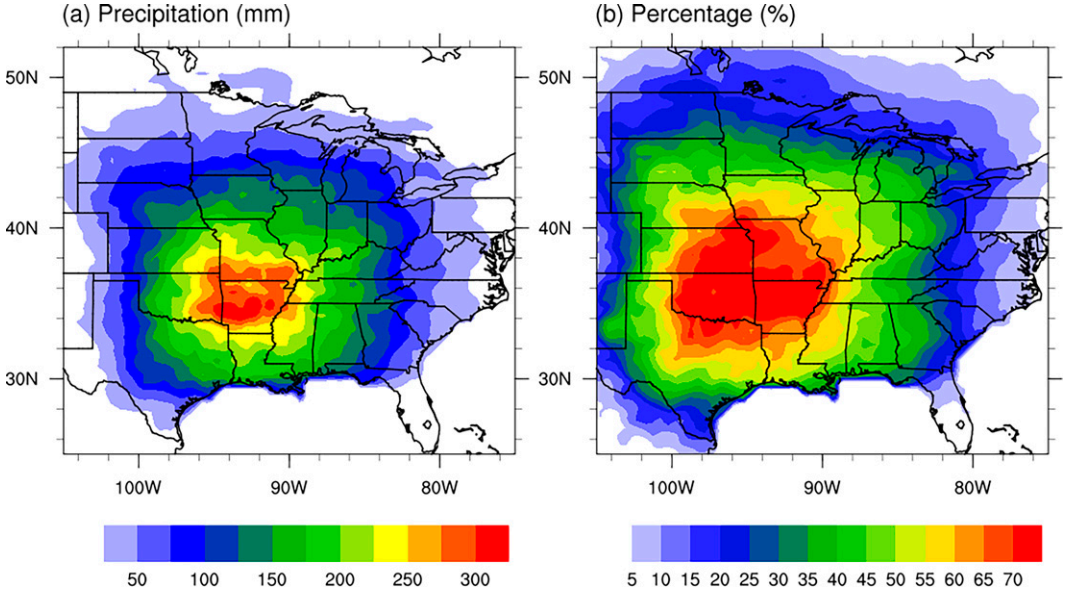


FIG. 4. (a) The climatological distribution of MCS-associated precipitation (mm) in boreal spring. (b) The percentage of MCS-associated precipitation to the total spring precipitation.

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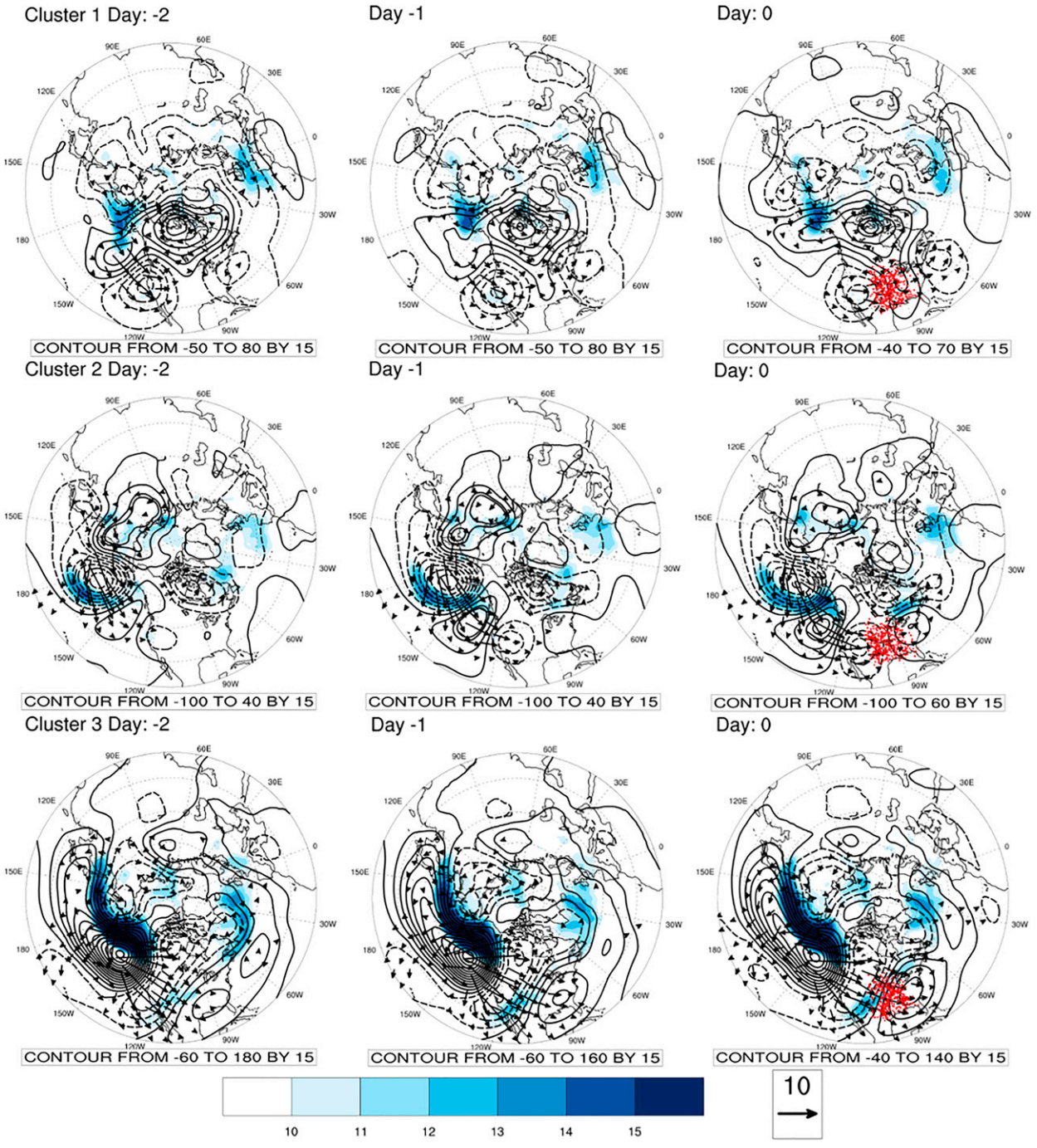


FIG. 5. The composite daily 500-hPa geopotential height anomalies (contour; gpm), 300-hPa wind speed anomalies (color shading; m s^{-1}), and 500-hPa wave activity flux (vector; m^2 s^{-2}) from day −2 to day 0 of MCS genesis in clusters 1, 2, and 3 obtained from the classification of large-scale forcing of MCSs. The contour has an interval of 15, with positive values in solid lines and negative values in dashed lines. Each row displays the evolution of the circulation anomalies for one cluster. Red dots on day 0 indicate all locations of MCS genesis in the corresponding cluster.

(Fig. 2a), we separate the entire detection domain into 0.25° × 0.25° pixels and count the number of MCS track points (the locations of MCS centers) falling into each pixel. The activity map (Fig. 2b) is generated from each MCS's intensity on all the associated cloud pixels (intensity at each pixel is defined as the difference between the warmer T_b threshold and the actual T_b of the pixel). Consistent with earlier studies (Feng et al. 2019; Haberlie and Ashley 2019),

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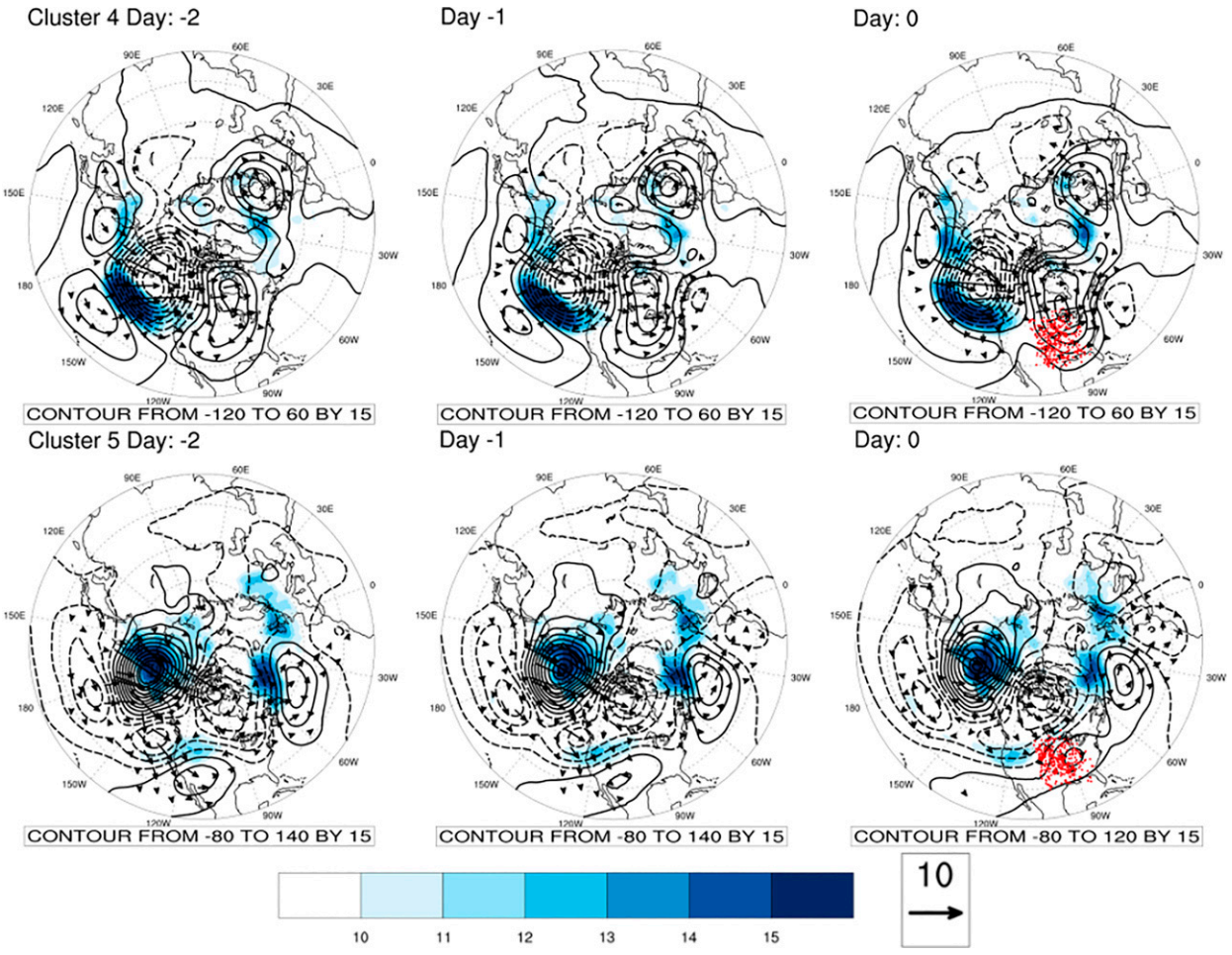


FIG. 5. As in the first three rows, but for cluster 4 and cluster 5.

MCSs are most frequent over the central and southern Great Plains (CGP and SGP) during spring. Both the MCS track density and activity peak near the joint center of the four states (Oklahoma, Kansas, Missouri, and Arkansas) in the SGP. A secondary peak in track density appears in the states of Louisiana, Mississippi, and Alabama. The absence of such a peak in the MCS activity map suggests that MCSs occurring in the U.S. Gulf Coast are typically weaker compared to those in the SGP.

The interannual variability (in terms of standard deviation) and the long-term trend of MCS genesis frequency are shown in Fig. 3. In obtaining these results, the number of MCS initiation points is added to the nearest 0.5° × 0.5° pixel over the entire domain. The SGP stands out as the region characterized by the largest interannual fluctuations of MCS occurrence, in addition to being the region of maximum MCS activity (Fig. 2). The SGP, part of the NGP, and Louisiana exhibit positive trends of MCS genesis while Iowa and Illinois display negative trends. The overall pattern supports earlier findings that the MCS frequency has significantly increased during the past 20 years (Feng et al. 2016). Figure 4 shows the MCS-associated precipitation over the United States and its contribution to the total spring precipitation. The spatial distribution of MCS precipitation resembles that of the MCS track density and activity (Fig. 2). MCSs account for more than 50% of the total precipitation in most parts of the Great Plains. In some regions this fraction exceeds 70%. Those results are consistent with conclusions drawn in earlier studies (e.g., Feng et al. 2019; Haberlie and Ashley 2019) and reaffirm the significance of MCSs in regulating the springtime water cycle over the Great Plains.

b. *Large-scale forcing of MCS genesis*

In this section, we compare the large-scale circulation patterns preceding MCS genesis in the five clusters identified (see section 2c for details). The clusters are ranked according to the number of MCSs they contain, with 371, 369, 335, 308, and 294 events in clusters 1–5, respectively. Each row in Fig. 5 displays the composite daily 500-hPa geopotential height anomalies and 300-hPa wind speed anomalies of one cluster from day −2 to day 0. The large-scale circulation patterns on day 0 are composited from the daily anomalies on the calendar date enclosing the moment 12 h prior to the MCS genesis. Since we are looking at the large-scale environments with

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eventual MCS genesis, 12 h represents a reasonable time scale for synoptic patterns to leave a footprint in MCS initialization. The composite 300-hPa wind speed anomalies are also added to each panel to illustrate the effect of upper-tropospheric jet streams in MCS initiation. The vectors in Fig. 5 represent the horizontal wave activity flux, which can be expressed as follows (Takaya and Nakamura 2001; Zhao et al. 2017a):

$$\mathbf{W} = \frac{p}{2|U|}\begin{bmatrix} U(\Psi_x'^2 - \Psi'\Psi'_{xx}) + V(\Psi'_x\Psi'_y - \Psi'\Psi'_{xy}) \\ U(\Psi'_x\Psi'_y - \Psi'\Psi'_{xy}) + V(\Psi_y'^2 - \Psi'\Psi'_{yy}) \end{bmatrix}, \quad (3)$$

where Ψ' stands for the perturbation streamfunction ($\Psi'_x = v'$, $\Psi'_y = -u'$); U and V are the daily climatological means of the zonal and meridional winds; $|U|$ is the magnitude of the wind (U, V); p is the pressure level normalized by $p_0 = 1000$ hPa.

In cluster 1 (top row of Fig. 5), a cutoff low appears near the U.S. West Coast on day −2. The system moves inland in the next two days and triggers MCSs (red dots in the right panel) over the Great Plains ahead of the cutoff low. Cluster 2 represents classic synoptic-scale disturbances originating from the Pacific storm track. A wave packet propagates downstream and intensifies across the North Pacific and North America within the time frame of several days. Subsequent analyses indicate that the MCS genesis frequency in cluster 2 is positively correlated with the index characterizing the Pacific–North American (PNA) teleconnection pattern (see Fig. 10). During the positive phase of PNA, the geopotential height features a negative anomaly over the North Pacific and a positive anomaly over the subtropics. This pattern shifts the Pacific jet southward (e.g., Lau 1988) and guides more storm track eddies propagating downstream toward the U.S. west coast, leading to more cluster 2 MCS events. The composite circulation pattern of cluster 3 is opposite to that of cluster 2 to some extent. A zonally elongated positive height anomaly replaces the dominant negative height anomaly found in cluster 2. The growth of a negative height anomaly with a large meridional extent ahead of the positive one is ultimately responsible for the MCS geneses, which are located slightly westward compared to the geneses seen in cluster 2. The circulation patterns of clusters 2 and 3 are therefore linked to different phases of storm track eddy development and are both characterized by pronounced downstream propagation of wave energy as indicated by wave activity fluxes. This contrasts the more "local growth" of the cutoff low in cluster 1, after its initial development triggered by energy dispersion from the upstream (top row, left panel of Fig. 5). In cluster 4, the circulation pattern appears more stagnant compared with those in clusters 1–3 and the center of a cutoff low is located near the Gulf of Alaska. The corresponding MCS geneses, interestingly, are seen behind a ridge situated over the northeastern United States, suggesting a weaker role played by large-scale forcing in the initiations of cluster 4 MCS events. Cluster 5 features a pronounced, zonally extended negative height anomaly occupying Canada and driving MCS formations over the central United States. Sustained by energy dispersion from the upstream height anomaly over the Bering Sea, the negative height anomaly remains semistationary during the three days evaluated.

In summary, the corresponding large-scale forcing in terms of the upper tropospheric trough–ridge pattern is directly relevant for MCS geneses in clusters 1, 2, 3, and 5. Among them, the anomalies of clusters 2, 3, and 5 are largely "remotely forced" while cluster 1 (also cluster 4) is characterized by more significant "local growth." The exact dynamical nature of these circulation anomalies remains to be delineated with further diagnostic and modeling analyses.

In addition to the potential curvature effect in driving large-scale ascent (therefore MCS genesis) discussed above, MCS geneses in clusters 3 and 5 are both attendant with positive 300-hPa wind speed anomalies to their west, suggesting the potential contribution from upper-level jet streams (Houze 2018; Maddox 1983). The exact impact of a jet stream on MCS genesis, however, depends on the precise location of an MCS relative to the jet—that is, left (right) to the entrance (exit) of the jet—which ultimately decides the jet stream forcing of upper-level divergence and vertical lifting. The clarification of this effect requires MCS genesis to be geographically well constrained in each cluster, which is not the case here. The scattering of genesis locations in individual clusters suggests the importance of additional local dynamic and thermodynamic factors in modulating the MCS initiation under a similar large-scale forcing. These factors are teased out in our second-level classification of MCS events falling into each of the five clusters.

c. *Local dynamic and thermodynamic forcing of MCS genesis*

A second-level hierarchical clustering applied to MCS track morphology further classifies the five clusters into 4, 4, 3, 5, and 4 subclusters (see section 2c for details). Several regions emerge as preferred locations of MCS activity including the SGP, the Midwest, the Ohio Valley, and the southeast United States. The inter-subcluster differences in local dynamic and thermodynamic forcing of MCSs largely determine the exact locations of MCS genesis under a similar large-scale forcing. To avoid redundancy, here we will evaluate these local factors in terms of composites based upon the subclusters with the largest number of MCS events within each of the main cluster. The conclusions drawn are in general applicable to other subclusters of less events. The local dynamic and thermodynamic factors in Figs. 6–8 are synthesized at precisely 6 h prior to the MCS initialization. This is also a reasonable time scale for capturing local and mesoscale impacts of the environment. We did not choose a longer time interval (e.g., 12 h) to minimize the potential impact of diurnal cycle on local factors.

Figure 6 shows the composite anomalies of sea level pressure and surface temperature for the selected five subclusters. Sea level pressure anomalies are included to detect prominent synoptic weather systems such as extratropical cyclones, "parent storms" of the MCSs. Large surface temperature gradients indicate the presence of synoptic-mesoscale fronts, a crucial mechanism of lifting. Except for cluster 4, all the subclusters are characterized by significant surface pressure anomalies with distinct surface lows present in clusters 2, 3,

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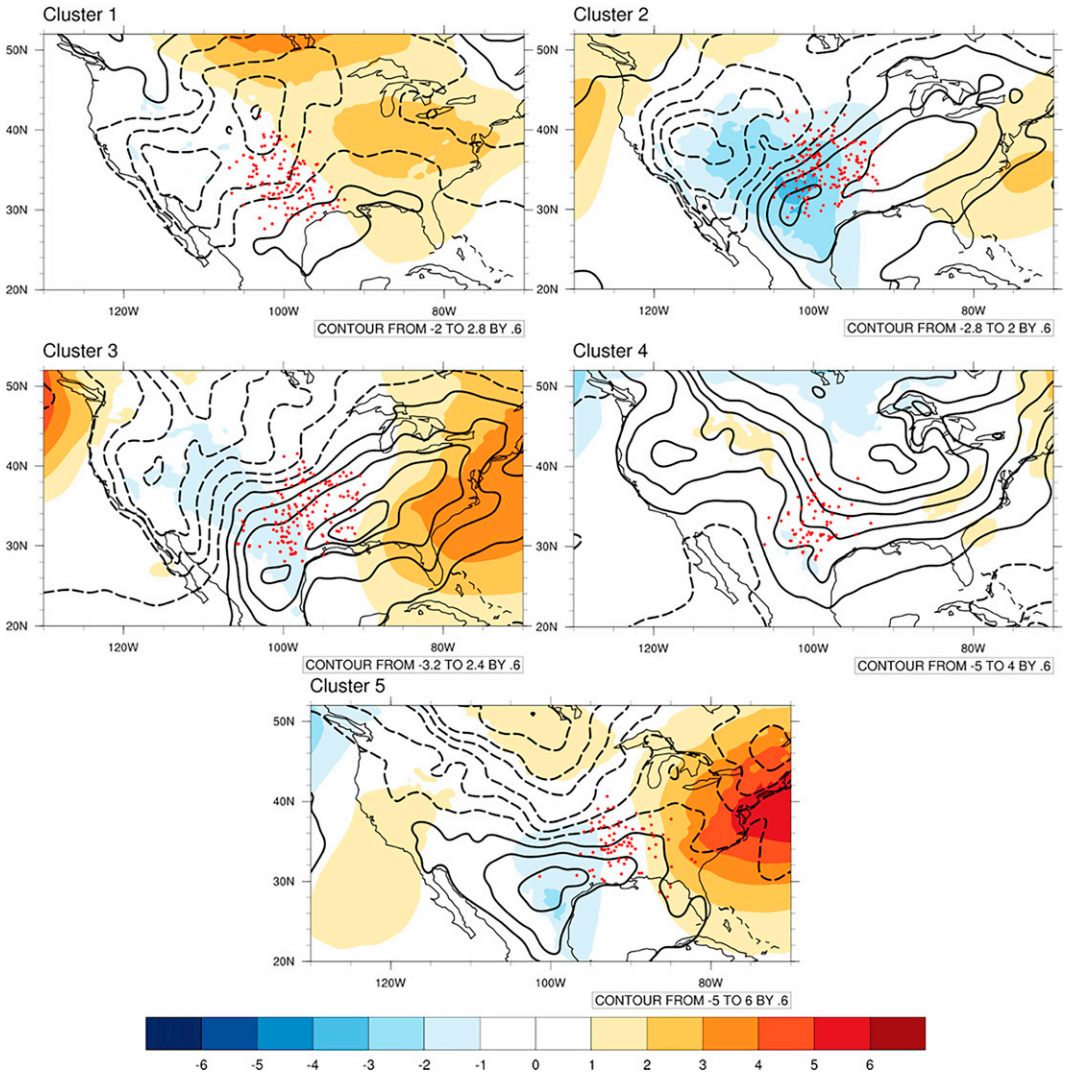


FIG. 6. The composite anomalies of surface temperature (contour; K) and sea level pressure (color shading; hPa) based upon MCSs in the subcluster with the largest number of events in each of the five main clusters. The contour has an interval of 0.6, with positive values in solid lines and negative values in dashed lines. Red dots indicate the genesis points of MCSs in each subcluster.

and 5. The weak large-scale forcing for cluster 4 MCS geneses seen in Fig. 5 is consistent with the absence of pronounced surface pressure anomalies found here. On the contrary, the strong upper-level forcing associated with troughs in clusters 2, 3, and 5 is well coupled with the development of surface cyclones, where MCS initiations tend to occur ahead of surface cold fronts in the warm sector. The absence of a distinct surface low in cluster 1 is likely the result of weaker baroclinic forcing associated with the zonally elongated cutoff low (top row of Fig. 5). The surface fronts characterized by the local tightening of temperature contours are present in all five clusters with varying intensities.

In addition to the surface variables, moisture transport vectors at 850 hPa may best capture the dynamic (low-level convergence) and thermodynamic effects of LLJs on MCS development (Fig. 7). What appears in all five subclusters in Fig. 7 is the prominent LLJs that carry moisture northward into regions of MCS genesis. These LLJs are partly associated with southerly winds ahead of a surface low and/or behind a surface high (Fig. 6), and are even present in cluster 4 with weak large-scale forcing. It is intriguing that the MCS genesis has a strong tendency to occur where the LLJ moisture transport converges (blue areas in Fig. 7). This correspondence indicates the key role of moisture transport in determining the location of MCS occurrence (e.g., Song et al. 2019).

In summary, large-forcing such as upper-tropospheric troughs sets up a broad envelope of anticipated MCS activity, and the presence of near-surface synoptic and mesoscale features including pressure anomalies, frontal lifting, and moisture convergence due to LLJs further determines the precise locations

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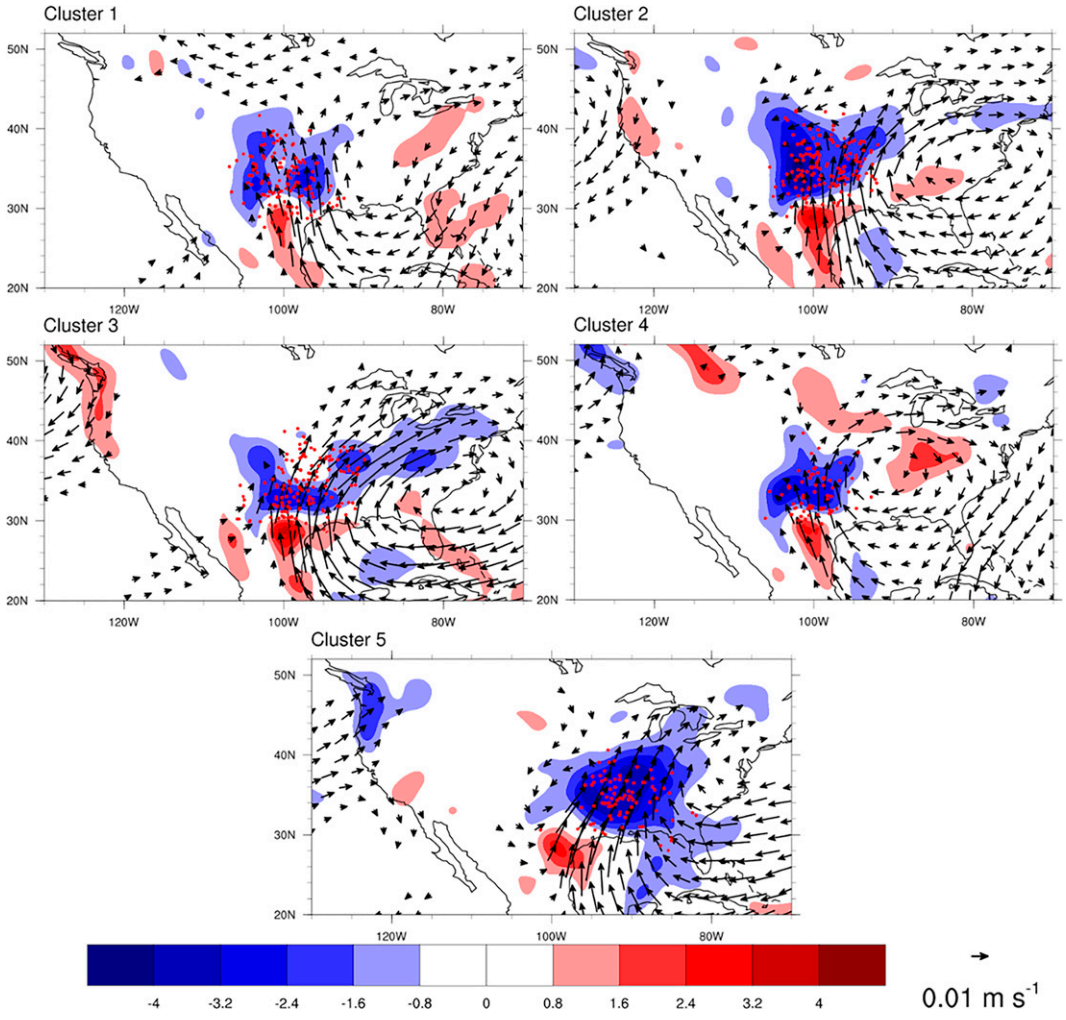


FIG. 7. The composite anomalies of 850-hPa moisture flux (vector; m s^{-1}) and moisture divergence (color shading; 10^{-8} s^{-1}) of the five subclusters in Fig. 6. Red dots indicate the genesis points of MCSs in each subcluster.

of MCS genesis. There are also occasions where large-scale forcing is weak and MCS initiations are linked to local frontal lifting and LLJs (e.g., cluster 4). It is important to note that the local dynamic and thermodynamic factors discussed here are intrinsically coupled with upper-tropospheric large-scale forcing (trough–ridge pairs and jet streams) and therefore an ultimate interpretation of MCS genesis and variability must come from a complete understanding of the life cycle dynamics and surface manifestations of large-scale circulation anomalies presented in Fig. 5.

To further understand the local thermodynamic forcing in a concise manner, we construct composites of CAPE (most unstable CAPE) and CIN 6 h prior to MCS genesis for the five subclusters discussed above. Note that 759 out of a total of 1677 MCS events (45%) analyzed are lacking CIN values at the initiation points (due to a mask applied to the dataset to cover data points where CIN $>$ 1000 J kg^{-1}). Those events are mostly accompanied with very small most unstable CAPE values (less than 50 J kg^{-1}) 6 h prior to the MCS initiation. This finding indicates that a significant portion of MCS events in boreal spring is likely "elevated convection" driven by potential instability. The key to realize potential instability is sustained lifting (often associated with fronts) that leads to condensation at the base of the potentially unstable layer. Sherwood (2000) found that such condensation may occur at the convective scale before the forced ascent converts potential instability into CAPE. Kreitzberg and Perkey (1976) described in detail how potential instability fuels convection with low CAPE values throughout the convective life cycle. More recently, McTaggart-Cowan et al. (2020) highlighted an important subset of MCS events that occur in low-CAPE environments where potential instability sustains moist convections. Our results suggest that approximately 45% of the springtime MCS events over the United States are related to potential instability. The genesis of these MCSs with very small CAPE values 6 h prior to MCS initiation are mostly found in the frontal regions of surface cyclones, confirming the importance of frontal lifting in releasing the potential instability.

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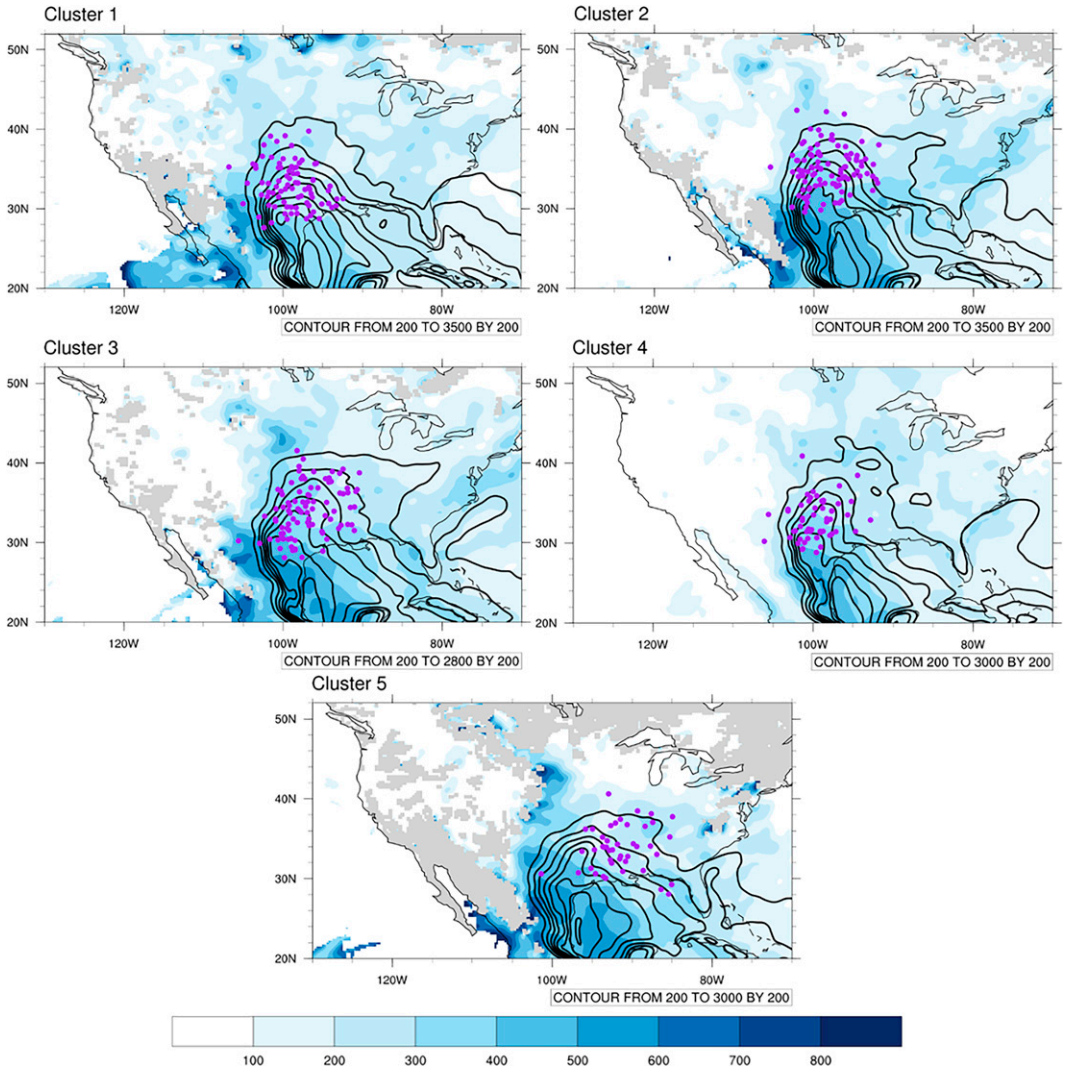


FIG. 8. CAPE (contour; J kg^{-1}; the interval is 200) and CIN (color shading; J kg^{-1}) composites of MCSs in the five subclusters. Purple dots indicate MCS genesis points in each subcluster.

The CAPE and CIN composites shown in Fig. 8 are thus based on the remaining 918 (55%) MCS events. A patch of large-CAPE values is seen over the Gulf of Mexico and extends to regions where MCSs are initiated in each subcluster. The importance of CAPE for MCS genesis reported earlier (e.g., Song et al. 2019) is supported by the collocations of large CAPE values and MCS initiations. The correlation between CIN and MCS genesis appears less clear even though many MCSs are initiated in an environment of relatively low CIN values. CAPE therefore seems to be a better predictor of MCSs compared to CIN. Ahijevych et al. (2016), using the random forest (RF) data mining technique, reached a similar conclusion by training an MCS prediction algorithm with a high-resolution numerical weather prediction model. Jirak and Cotton (2007), however, came to a different conclusion and ranked CIN ahead of CAPE in the predicting skill of MCSs based on the Heidke skill score (HSS). It is worth mentioning that the evaluation of Jirak and Cotton (2007) is to distinguish MCSs from the common "widespread convections." They pointed out that CIN in an MCS precursory environment is much larger than that in an environment of common convection, since larger CIN may produce a modest cap, inhibiting convective energy release and instead leading to the development of explosive convection and MCSs.

Figure 9a shows the occurrence frequency of MCSs in boreal spring as a function of CAPE and CIN. The largest numbers of MCS events are found with CAPE (CIN) around 300 (100) J kg^{-1}. Most MCSs occur with CIN below 300 J kg^{-1} and CAPE up to 1000 J kg^{-1}. Note that even in these cases with relatively large CAPE values, potential instability may still contribute to MCS genesis. This explains why the initiation of some MCS events does not require large CAPE values in spring when frontal activities (thus lifting) are much more prominent compared to those in summer. Figure 9b displays the MCS genesis

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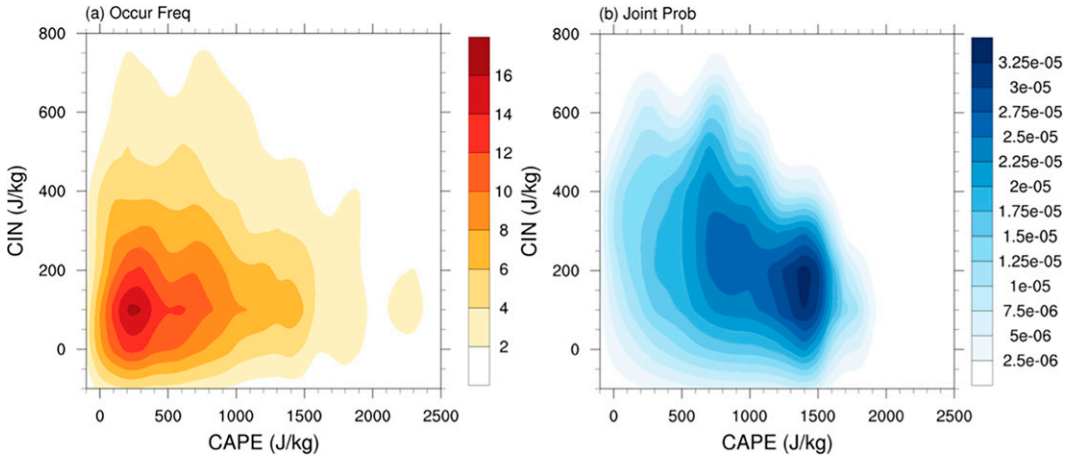


FIG. 9. (a) The occurrence frequency of MCS as a function of CAPE and CIN. (b) The genesis probability of MCSs conditioned on CAPE and CIN in boreal spring.

probability conditioned on CAPE and CIN values. It is obtained by dividing the MCS frequency (Fig. 9a) with the occurrence frequency of the corresponding pair of CAPE and CIN among all the hourly values in the ERA dataset within the spatial and temporal MCS detection domain (30°–50°N, 80°–105°W; the 2000–20 springtime). The highest genesis probability is found in an environment characterized by CAPE (CIN) around 1400 (150) J kg^{-1}. The much greater CAPE value (1400 J kg^{-1}) compared to that in Fig. 9a and the well-constrained maximum probability by CAPE demonstrate the significance of large CAPE for MCS genesis and suggest the value of including CAPE as a predictor in MCS forecasting.

d. Variability and trend of MCS genesis

The delineation of multiscale forcing of MCSs in earlier sections allows us to understand better the origin of interannual variability and long-term trends in MCS genesis. Specifically, quasi-stationary teleconnection patterns give rise to prolonged anomalous weather situations over certain regions by changing the paths of migratory, high-frequency synoptic systems. Figure 10 shows the time series of the 2000–20 springtime MCS frequency for each of the five clusters obtained through the classification of large-scale forcing. Trend analysis and correlation with teleconnection indices (obtained by taking the springtime average of NOAA Climate Prediction Center teleconnection indices) are also performed. Cluster 2 is the only one exhibiting a significant upward trend and is also positively correlated with the PNA index at the 0.01 level of statistical significance (see the numbers in the box next to the plot in Fig. 10). Cluster 3 is negatively correlated with the PNA, consistent with the findings that clusters 2 and 3 represent different phases of storm track eddy development characterized by nearly opposite circulation anomalies across the Pacific–North America sector (Fig. 5). Figure 11 replicates Fig. 3 with MCS events from cluster 2 only. The general consistency between Figs. 3 and 11 indicates that MCS events associated with cluster 2 type of large-scale forcing account for majority of the interannual variations of springtime MCS over the United States and also lead to the overall positive trend of MCS number in recent decades.

Figure 12 shows the spring-mean 500-hPa geopotential height and SST regressed onto the detrended springtime MCS genesis frequency of cluster 2. The distribution of the slope coefficients of geopotential height in Fig. 12a resembles that of the positive PNA, consistent with the positive correlation between cluster 2 and the PNA found earlier. The overall SST distribution over the North Pacific in Fig. 12b bears great similarity to that of a positive phase of the Pacific decadal oscillation (PDO) (Mantua and Hare 2002), that is, negative anomalies over the North Pacific and warm anomalies along the west coast of North America and in the equatorial central and eastern Pacific. This similarity is also confirmed by the significant positive correlation between cluster 2 and the PDO index (Fig. 10). It is well known that the PDO has exhibited a significant positive trend over the past two decades (Zhang et al. 2018; Zhou et al. 2017) and the North Pacific SST has a distinct footprint in the U.S. warm season rainfall (Zhao et al. 2017b). The findings here suggest that the PDO is likely the main climate mode that modulates springtime MCS frequency across decadal time scales and is directly responsible for the positive MCS trend seen over a short period of 20 years.

4. Concluding remarks

Using clustering analysis, we conduct a systematic and hierarchical dissection of the multiscale forcing of MCS genesis over the United States in boreal spring for the period 2000–20. MCS events are first detected through a "detect and spread" approach. A climatological assessment of these events indicates that MCSs are most active over the CGP and SGP in terms of both occurrence frequency and intensity and that they account for more than half of the springtime precipitation over the Great Plains. Classification of spatiotemporal features of upper-tropospheric height anomalies leading to MCS genesis over the United States reveals five distinct large-scale forcing patterns (clusters) associated with different configurations

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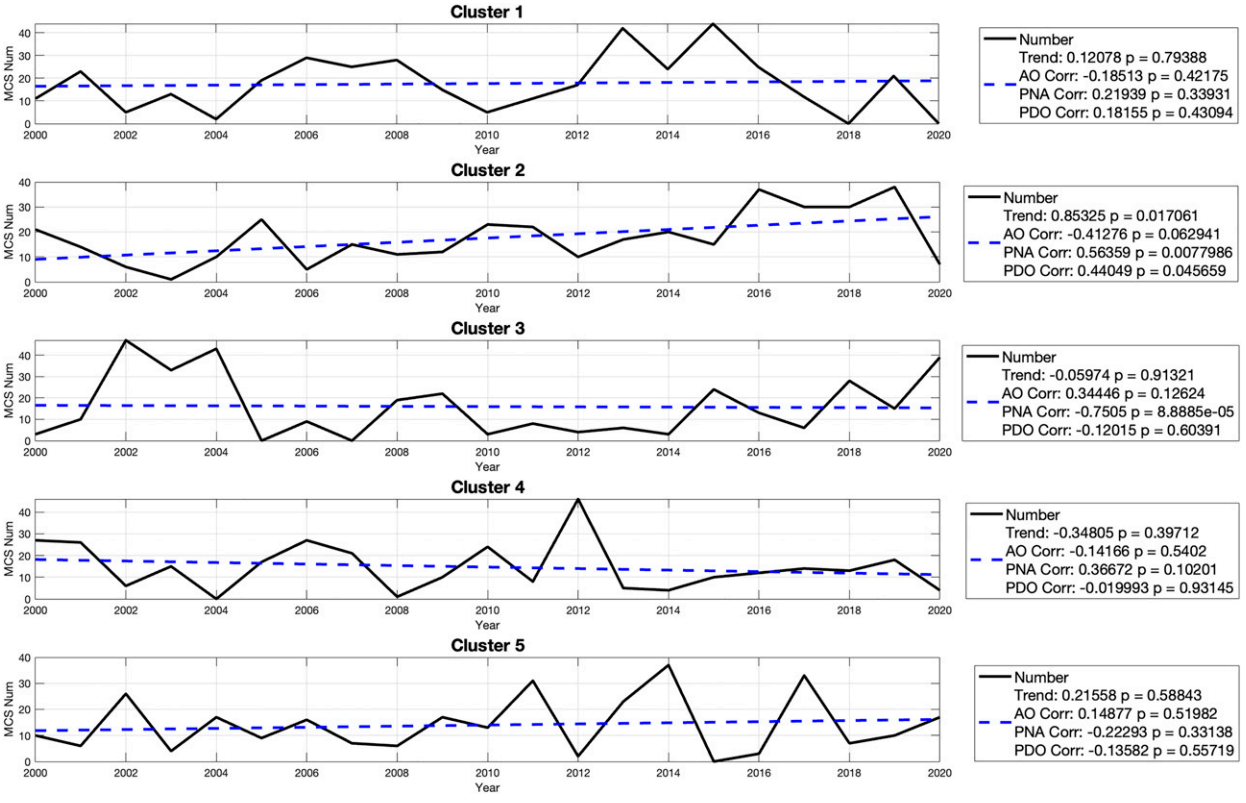


FIG. 10. Time series of the spring MCS genesis frequency in the period 2000–20 for the five main clusters identified (black solid lines). Blue dashed lines indicate linear trends. Correlations with prominent atmospheric teleconnection indices are given in the boxes next to the plots.

and intensities of 500-hPa troughs and ridges. The forcing patterns fall into two broad categories in terms of their dynamical origins: "remotely forced" and "local growth". The "remotely forced" category includes upper-tropospheric troughs associated with downstream propagation of storm track eddies (clusters 2 and 3) and energy dispersion from the Bering Sea (cluster 5). The "local growth" category includes the development of a cutoff low over the western United States (cluster 1) and a stagnant trough–ridge pattern over the Gulf of Alaska and North America (cluster 4). The upper-tropospheric troughs create an envelope downstream where large-scale lifting and MCS initiations tend to occur except for cluster 4, in which case upper-tropospheric circulation anomalies do not appear to be directly responsible for MCS genesis. The direct effects of upper-tropospheric jet streams are generally weak and present only in clusters 3 and 5.

Among all the five clusters, MCSs associated with cluster 2 type large-scale forcing (i.e., downstream propagation of wave

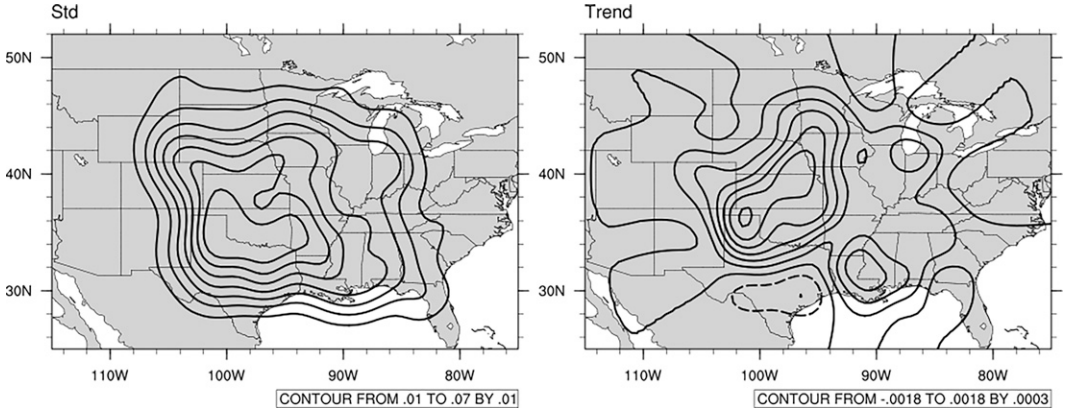


FIG. 11. As in Fig. 3, but for MCSs in cluster 2 only.

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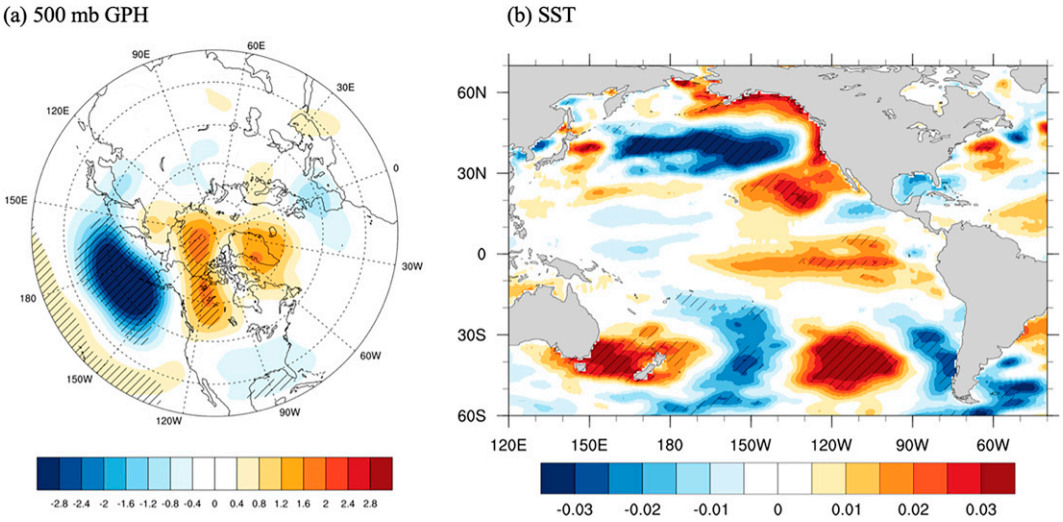


FIG. 12. (a) The spring-mean 500-hPa geopotential height (gpm) and (b) the sea surface temperature (K) regressed onto the detrended MCS genesis frequency of cluster 2 for the period 2000–20.

packets from the Pacific storm track) account for the majority of the MCS interannual variations over the United States and are directly responsible for the positive trend of the United States. MCSs observed in the past two decades. Cluster 2 is also correlated significantly with the PNA teleconnection and the PDO. As the PDO index has been consistently in a positive phase in most recent decades, our findings suggest that the PDO is likely the main climate mode that modulates the springtime MCS frequency across decadal time scales and ultimately responsible for the positive MCS trend seen over a short period of 20 years.

A second-level classification of MCS events based upon MCS track geophysical locations separates each of the five main clusters into multiple subclusters. Composites of sea level pressure, surface temperature, and lower tropospheric moisture flux for these subclusters reveal the crucial roles played by local dynamic and thermodynamic factors in determining the precise locations of MCS genesis within the envelope created by upper-tropospheric large-scale forcing. MCSs are found to often occur near surface fronts in the warm sector of low pressure systems, suggesting the importance of frontal lifting for convection initiation. Low-level (moisture) convergence associated with LLJs turns out to be a key element in determining MCS genesis locations. About 45% of the springtime MCSs are triggered by the release of "elevated" potential instability through frontal lifting. The remaining 55% of the MCS events occur most commonly when CAPE is around 300 J kg^{-1} and CIN is around 100 J kg^{-1}. The combination of 1400 J kg^{-1} CAPE and 150 J kg^{-1} CIN gives the highest probability of MCS genesis, providing strong implication for MCS forecasting. Ongoing and future work includes a further delineation of the dynamical nature of MCS large-scale forcing identified here and more diagnostic and modeling analyses to understand the coupling of large-scale forcing and local synoptic-mesoscale features such as surface cyclones, fronts and LLJs. Another potential application of the multiscale forcing documented in this study is to trace the origins of MCS frequency biases in global climate models.

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Data availability statement. The global merged geostationary satellite infrared brightness temperature dataset is produced by the NOAA Climate Prediction Center and archived at the NASA Goddard Earth Sciences Data and Information Services Center (GES DISC) at https://disc.gsfc.nasa.gov/datasets/GPM_MERGIR_1/summary. The atmospheric variables on the single level and all pressure levels come from the fifth generation of ECMWF reanalysis dataset (ERA5) and are available on the Climate Data Store (https://cds.climate.copernicus.eu/#!/search?text=ERA5&type=dataset). The CPC global gridded daily precipitation and the COBE-SST dataset can be obtained from the website of gridded data products by NOAA Physical Sciences Laboratory (https://psl.noaa.gov/data/gridded/).

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